

Part A · Multiple Choice

2 points each

1. **a) Principle of Superposition**

p. 47 — first developed by Nicholas Steno. Original horizontality says the layers started out flat; lateral continuity says they extend sideways until they thin out.

2. **a) “the present is the key to the past”**

p. 47 — Hutton realized geologic processes are slow, which gave geologists a basis for interpreting ancient rocks using processes we can watch today.

3. **d) Phanerozoic**

p. 45 — the first three eons together represent almost 90% of Earth’s history. Phanerozoic rocks are the most common on Earth and contain familiar fossils.

4. **b) Paleozoic, Mesozoic, Cenozoic**

p. 45 — their names mean “early life,” “middle life,” and “new life,” which is a reliable way to remember the order.

5. **c) 57,300 years**

p. 50 — beyond about ten half-lives, so little parent isotope remains that accurate chemical analysis is no longer possible.

6. **b) 25% parent and 75% daughter**

p. 49 — the daughter-to-parent ratio is 0 at the start, 1 after one half-life, and 3 after two half-lives.

Part B · True or False

1 point each

7. **FALSE**

Relative dating establishes only whether one rock or event is older or younger than another. Numerical ages require absolute (radio-isotopic) dating, which was not available until radioactivity was discovered in the late 1800s. (pp. 47–48)

8. **TRUE**

p. 44 — Smith discovered that fossils could be used to correlate rocks of the same age, and used this to publish the first geological map of England and Wales in 1815. He is known as the “Father of English Geology.”

Part C · Short Answer

3 points each

9. **A mass spectrometer.**

Rubric: 1 pt for naming the mass spectrometer. 2 pts for the process: beams of the isotopes are passed through a magnetic field, and because the path of the heavier isotope is deflected less, the beams strike a sensor at different places. The intensity of each beam gives the amount of parent and daughter product, and the *ratio* of daughter to parent yields the age. Credit the key idea — age comes from the parent-to-daughter ratio, not from either amount alone. (pp. 48–49)

10. Geology requires the understanding and application of all the other sciences, and it adds the dimension of deep time.

Rubric: 2 pts for explaining that geology draws on physics, chemistry, biology, mathematics, and astronomy together (naming at least three earns full credit). 1 pt for identifying the extra dimension as *time* — deep time, billions of years of it. Strong answers note that geologists usually observe the *results* of processes that happened thousands to billions of years ago, at rates of only millimeters to centimeters per year. (p. 40)